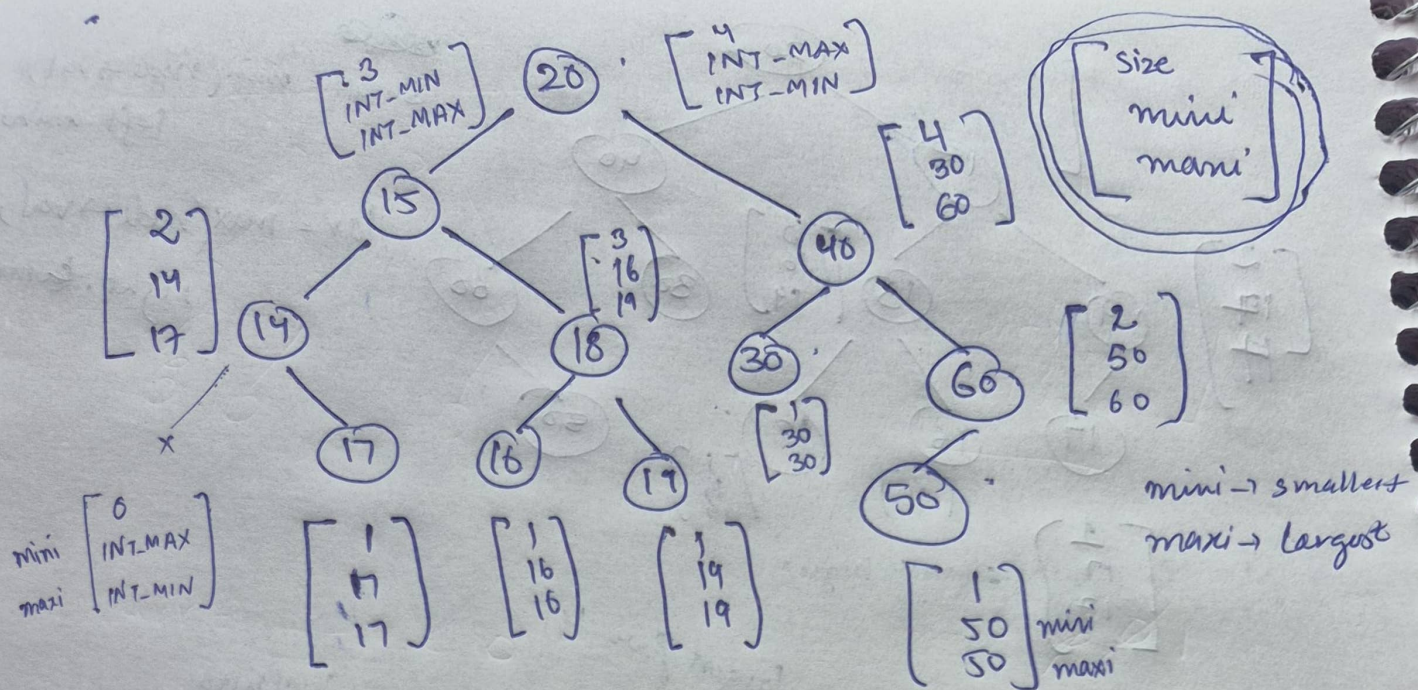




- mini = $\min(\text{node} \rightarrow \text{val}, \text{left.min})$
maxi = $\max(\text{node} \rightarrow \text{val}, \text{right.max})$ } Normal range
- Initially, mini = INT-MAX
maxi = INT-MIN } ~~also for~~ NULL too
- When Validation fails, mini = INT-MIN
maxi = INT-MAX

Validation
condition $\Rightarrow$

largest < node $\rightarrow$ val < smallest
from left from right
subtree subtree

① $\text{INT_MIN} < 14 < 17$ ✓

② $16 < 18 < 19$ ✓

③ $17 \nless 15 < 16$ ✗

④ $50 < 60 < \text{INT_MAX}$ ✓

⑤ $30 < 40 < 50$ ✓

⑥ $\text{INT_MAX} \nless 20 < 30$ ✗

$$\begin{bmatrix} 3 \\ \text{INT_MAX} \\ \text{INT_MAX} \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ \text{INT_MAX} \\ \text{INT_MIN} \end{bmatrix}$$